C-JDBC Developer's Guide

Emmanuel Cecchet

Julie Marguerite

Mathieu Peltier

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1. Getting the source code

1.1. Source distribution

The C-JDBC source distribution can be downloaded from the C-JDBC's Web site (http://c-jdbc.objectweb.org/). It includes the source code of both C-JDBC driver and controller as well as the test suite and all tools.

The following three formats are available (where `x.y` is the C-JDBC release number):

- `c-jdbc-x.y-src-installer.jar`: Java™ graphical installer (powered by IzPack (http://www.izforge.com/izpack/)).
- `c-jdbc-x.y-src.tar.gz`: source distribution for the Unix platforms users.
- `c-jdbc-x.y-src.zip`: source distribution for the Windows platforms users.

The easiest way to install the source distribution is to use the Java graphical installer[1]. If you want to use the other distribution formats, just uncompress the downloaded file in the directory of your choice.

> **Note:** Nightly builds releases are also available for download. These releases are regenerated each night at 23:45 PM CEST.

1.2. CVS access

The whole C-JDBC codebase can be checked out through anonymous access using any CVS (Concurrent Versions System) client. The CVS server to use is `cvs.c-jdbc.forge.objectweb.org` (in `pserver` mode). The C-JDBC repository is named `/cvsroot/c-jdbc`. Unix users can use for example the following commands:

```
bash> cvs -d:pserver:anonymous@cvs.c-jdbc.forge.objectweb.org:/cvsroot/c-jdbc login
bash> cvs -d:pserver:anonymous@cvs.c-jdbc.forge.objectweb.org:/cvsroot/c-jdbc co .
```

When prompted for a password, just hit enter.

The available modules are:

- `c-jdbc`: used to get the C-JDBC codebase.
- `web`: used to get the C-JDBC Web site.
- `.`: used to get the whole C-JDBC repository.

> **Note:** The latest snapshot of the CVS repository is also available for download. The files to download are: `c-jdbc-current-cvs.tar.gz` or `c-jdbc-current-cvs.zip`. These are automatically regenerated each night at 00:00 AM CEST.

1.3. Distribution organization

C-JDBC source code comes with an Ant (http://ant.apache.org/) build file. You will need **ant** (release 1.5 or greater) to build C-JDBC. You can get Ant and its documentation from the Apache's Web site (http://ant.apache.org/).

Tunable build properties are set in the `build.properties` file in the root directory of the distribution. The only property you usually have to set is the installation directory (property `install.dir`) where you want to install the distribution once it is compiled.

ant clean dist will build both source and binary distribution files in the `build` directory.

ant clean install will build a C-JDBC binary distribution and install it in the directory specified by the `install.dir` property defined in the `build.properties` file.

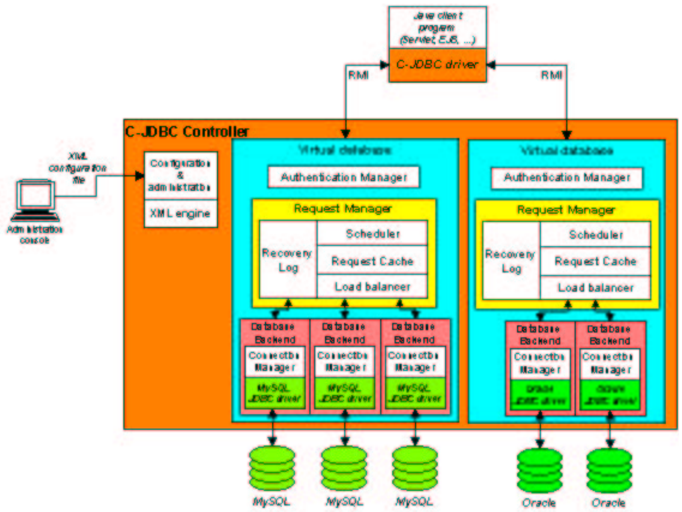
You can get an extensive list of the ant targets using the **ant -projecthelp** command. Here is an overview of the available targets:

- **clean**: deletes the `${build}` and `${dist}` directory trees and emacs backup files.
- **compile**: compiles the C-JDBC driver, controller and request player (the **compile-controller**, **compile-driver**, **compile-requestplayer** and **compile-tests** sub targets compile only the C-JDBC controller, driver, request player and JUnit (http://www.junit.org/) test suite, respectively).
- **dist**: creates binary and source C-JDBC distributions (`tar.gz`, `zip` formats and Java graphical installer powered by IzPack (http://www.izforge.com/izpack/)). The **dist-bin** and **dist-src** sub targets create only binary or source distributions, respectively.
- **doc**: generates documentation in various format from DocBook (http://www.docbook.org/) sources. Warning, this target requires the following tools: **make**, **LaTeX**, **jade**, **jadetex**, **dvips**, **dvipdf** and should only work under Unix platforms. Please refer to the specific documentation of your system to known how to install them.
- **doccheck**: checks the Javadoc syntax and produces detailed errors report, using the Sun Doc Check Utilities (http://java.sun.com/j2se/javadoc/doccheck/).
- **install**: installs C-JDBC distribution in the directory specified by `${install.dir}`
- **jar**: creates JAR files for C-JDBC driver, controller and request player (the **jar-controller**, **jar-driver** and **jar-requestplayer** sub targets create only C-JDBC controller, driver and request player JAR files, respectively).
- **javadoc**: generates Javadoc documentation.
- **test**: runs all JUnit tests (warning `junit.jar` and `xalan.jar` must be found in `${ANT_HOME}/lib`).
- **test-gui**: launches the GUI test runner of JUnit.
- **test-unit**: runs one JUnit test specified by `-Dtest.class=<class name>` (warning `junit.jar` and `xalan.jar` must be found in `${ANT_HOME}/lib`).

2. Design overview

To be documented.

Figure 1. C-JDBC controller design overview example



3. Architecture overview

To be documented.

4. Code conventions

C-JDBC uses the Sun’s Java standard code conventions (http://java.sun.com/docs/codeconv/html/CodeConvTOC.doc.html) with little modifications (for example, braces in statements, ...). For more details, please see the source code itself.

Eclipse (http://www.eclipse.org/)[2] users can checkout from the C-JDBC CVS repository a property file containing all the Eclipse Code Formatter settings to suite with the C-JDBC code conventions. This file is named `.codeFormatter.properties` in the root of the CVS repository. It can not be imported in Eclipse (version 2.1): you have to change manually the Eclipse Code Formatter preferences.

The CVS repository also contains a configuration file for the Checkstyle (http://checkstyle.sourceforge.net/) Eclipse plugin. This plugin allows you to check code conventions and Javadoc automatically in Eclipse and provides the user with many facilities to correct errors. This property file is named `.checkstyle.properties` in the root of the CVS repository and can be imported to modify automatically the Checkstyle preferences (open “Window” menu, then choose “Preferences” and click to “Checkstyle”)[3].

Notes

1. . A Java Virtual Machine™ is of course needed in this case.
2. Eclipse is a popular open source IDE for Java.
3. The CVS repository contains also standard Eclipse project files that can be used to manage a `c-jdbc` project (`.project` and `.classpath`).